Relief Valve Sizing

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Learning Outcomes

- Perform preliminary sizing calculations of a relief valve for gas flow scenarios.

Reading

- Foundations of Spiritual and Physical Safety: with Chemical Processes; Section VII.3.1.

In the orifice flow lecture, we reviewed estimated flows through an orifice for gases, liquids, and two-phase flow. Those equations can be used to size a relief valve to prevent the pressure from exceeding a certain value. Solving the mass or mole balance equations in such scenarios can often be completed in a straight forward way.

Important parameters for estimating the internal pressure to ensure the maximum pressure allowed isn't exceeded include:

- the generation rate of the gases from the failure mode that need to be vented
- the size of the relief valve

The American Petroleum Institute publishes the API 520 which is a standard for sizing and selection of relief devices. The newest edition is the 10th edition and it is 509 dollars on their webstore: https://www.apiwebstore.org/standards/520__Part_1 . Although I don't have the most recent edition, an estimate for the needed area for gas flow is by the below equation.

0.1 API 520 Equation in SI units

area in mm for this flow rate in $\frac{kg}{hr}$

temp

compressibility factor

$Z = \frac{PV}{RT} = 1$

$$A = \frac{W}{C \cdot K_d \cdot P_1 \cdot K_b \cdot K_c} \sqrt{\frac{T \cdot Z}{M}} \tag{1}$$

k, γ $\quad 0.9 \quad 1$

where A is the vent area in mm, W is the mass flow through the vent in kg/hr, P_1 is the upstream relieving pressure (set pressure + overpressure + atmospheric pressure) in absolute pressure in kPa, T is the temperature in K, Z is the compressibility of the gases (1), M is the molecular weight of the gases in gm/mol, C is a value incorporating the heat capacity ratio and unit conversion and has a value here of 0.027. K_d is the coefficient of discharge equal

to 0.9. K_b and K_c are other coefficients depending on downstream resistance or other configurations and are equal to 1 in this case.

0.2 Gas Flow Through an Orifice (Theoretical)

Reproduced here is estimated the flow of a gas through an orifice from the orifice flow lecture. It is also derived from an energy balance assuming adiabatic and frictionless flow. The flow is characterized by the pressure ratio, P_o/P_{atm} and associated Mach number, Ma. P_o is the absolute pressure of the volume. The Mach number can be found from:

Mach #

$$Ma = \min\left(1, \sqrt{\frac{2}{\gamma - 1}\left(\left(\frac{P_o}{P_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right)}\right) \tag{2}$$

where γ is the ratio of specific heats (1.4 for air). The flow is choked when $Ma = 1$ and unchoked when $Ma < 1$.

The molar flow rate is given by:

gas molar flow rate

$$\dot{n} = P_o A C_d \sqrt{\frac{\gamma}{RTM_w}} Ma \left[1 + \frac{(\gamma - 1)}{2} Ma^2\right]^{\frac{\gamma+1}{2-2\gamma}} \tag{3}$$

solve

where R is the gas constant and T is the temperature, P_o is the absolute pressure inside the vessel, C_d is the discharge coefficient (typically about 0.9). If you use all SI units (Pascal, meters, kilograms, mole: so pressure in pascal, area in meters, R in J/mol/K, and Mw in kg/mol), you'll get mole per second for $\dot{n}$. See Perry's Handbook for Chemical Engineers Equation 6-118 for more information.

Relief valve sizing in some instances does required verification with actual flow per the AMSE Boiler and Pressure Vessel Code.

> **Note**
> See in-class examples
>
> Download a pdf of the example using freeform here: supportfiles/GasFlowExample311.pdf

1 In Class Example: Venting CO2 from 2L Bottle

A two liter bottle is half filed with water. There is a relief valve on the top designed to relieve at 60 psig with a flow of 61 scfm. What is the flow rate of CO2 through the relieve per the above equations?

61 scfm

```
import numpy as np

#fist find the Mach number
gamma = 1.3 #specific heat capacity ratio for CO2, estimate
P1 = 60 + 12.5 #initial pressure, estimate, psia
Patm = 12.5 # atmospheric pressure, psia
Ma = min(1, np.sqrt(2/(gamma -1)*((P1/Patm)**((gamma -1)/gamma) -1))) #Mach number

Ma

1

#now calculate the molar flow rate
R = 8.314 # gas constant, J/(mol*K)
Temp = 295 # temperature, K
```

```
Area = np.pi/4*(1/4*0.0254)**2 # area of the pipe, m^2
Cd = 0.9 # discharge coefficient
Mw = 0.04401 # molar mass of CO2, kg/mol
# molar flow rate, mol/s
ndot = P1*6894.76*Area*Cd*np.sqrt(gamma/(R*Temp*Mw))*Ma*(1+(gamma -1)/2*Ma**2)**((gamma+1)/(2 -2*gamma))
print(f'The molar flow rate is {ndot:0.2f} mol/s')
```

orifice

```
The molar flow rate is 0.92 mol/s
```

```
#now convert that molar flow rate into scfm, PV = nRT
Tstandard = 298.15 # standard temperature, K
Pstandard = 101325 # standard pressure, Pa
scmps = ndot*R*Tstandard/Pstandard #units of m^3/s
scfm = scmps*60*(3.281**3)
print(f'The flow rate in scfm is {scfm:0.1f}')
```

```
The flow rate in scfm is 47.4
```

Why is this number different than the reported flow rate of CO2 through the relief valve? Could the diameter of the relief be slightly larger?

How would you get the generation rate of CO2 from the submlimation of dry ice?

$$Q = hA(T_{water} - T_{ice})mdot = Q/\Delta H_{vap} \tag{4}$$

Does area (A) change? How would you get the heat transfer coefficient (h)?

2 In Class Example 2: More venting from a CO2 bottle

See text, in-class notes, and the python code here: Calcs.ipynb (after the Chapter 7 Heading)

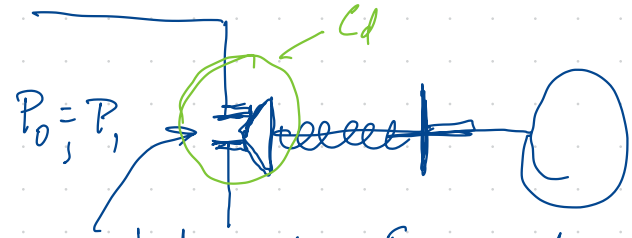
Action Items

1. Determine the size of the vent needed for a 50 cubic meter tank filling with water at 5 kg/sec, assuming a MAWP of 1.5 barg and a discharge coefficient of 0.95.
2. Inside of a 100 cubic meter tank (half full of liquid), nitrogen is being generated at a constant rate of 1 mol per second. Determine the minimum diameter of a spring operated relief valve to prevent the over pressurization of the tank. The tank has an MAWP of 3.5 barg, and the nitrogen is generated at $40°$C. Assume the valve vents to atmosphere and use a standard 10% overpressure for your calculations. Please calculate the recommended vent area (in mm) two different ways:

- Using API 520 Equation given below
- Using the orifice flow equations given in the Orifice Flow lecture, specifically the equation used for flow of gases through an orifice, equations also given below.
- Compare and comment on the two answers.

- Read problem

Example Problem 1 from Lecture



what is the flow rate through the orifice or PRV? in scfm

$Ma = \frac{>1}{\quad} = 1$

$\dot{n} = \frac{0.92 \text{ mol/sec}}{\text{min}}$ → standard cubic feet per minute

mol → ft³

$V = \frac{nRT}{P}$ ← 298 K, 1 atm

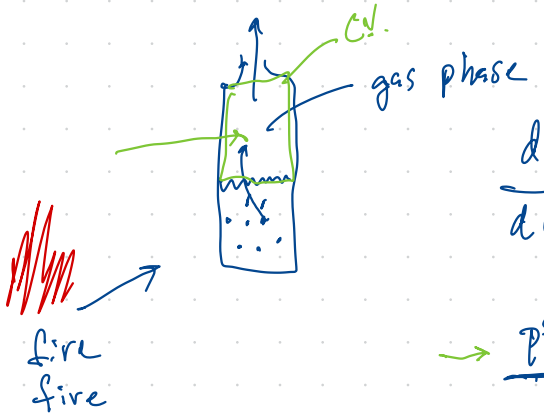
sfm: 47.4 ≠ 61 scfm

gases - 0.9 liquid - 0.65 → Cd

$\gamma = 1.4$ for air

$Cd < 1$ ~1

Example Problem From the Book



$$\left.\frac{dn}{dt}\right|_{C.V.} = \dot{n}^{CO_2}_{des} + \dot{n}^{H_2O}_{vap}$$

desorbing

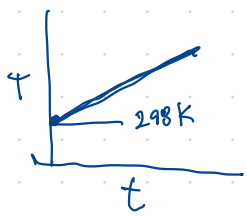
$$p^{sat} = e^{A - B/T}$$

August

disolved

$$n_{dis} = P \cdot m_L \cdot H$$

$$H = H_0 \, e^{\boxed{-A\left(\frac{1}{T} - \frac{1}{T_0}\right)}} \quad \leftarrow f(T)$$



chain rule

$$-\frac{dn_{dis}}{dt} = \dot{n}_{des} = \boxed{P m_L \cdot \frac{dH}{dt} + H m_L \frac{dP}{dt}} + \cancel{P H \frac{dm_L}{dt}} \sim 0$$

$$\frac{dH}{dt} = H \frac{A}{T^2} \frac{dT}{dt}$$

$$p^{sat} = e^{A - B/T} \rightarrow \text{derivative} \quad e^{A - B/T} \cdot \frac{d}{dt}\left[A - \frac{B}{T}\right]$$

$$\frac{d}{dt}\left[-BT^{-1}\right]$$

$$BT^{-2}\frac{dT}{dt}$$

$$\Delta P \leftarrow \boxed{\frac{dp^{sat}}{dt} = p^{sat} \frac{B}{T^2} \frac{dT}{dt}}$$

$$\underset{\text{constant}}{PV} = nRT$$

$$\frac{dP}{dt} \cdot V = RT \frac{dn}{dt} + nR \frac{dT}{dt}$$

$$\frac{d}{dt}\left[-BT^{-1}\right]$$

$$-T^{-1}\cancel{\frac{dB}{dt}} + -B \frac{dT}{dt}$$

$$\frac{d}{dt}\left[-BT^{-1}\right]$$

$$-1\, T^{-2}\frac{dT}{dt}$$

$$-\frac{B}{T^2} \cdot \frac{dT}{dt}$$

Ideal Gas Law Derivatives

$$V\frac{dP}{dt} = nR\frac{dT}{dt} + RT\frac{dn}{dt}$$

$$PV = nRT$$

constant V $\quad \frac{dV}{dt} = 0$

$$\frac{dP}{dt} = \boxed{\frac{nR}{VP}}\frac{dT}{dt} + \boxed{\frac{RT}{PV}}\frac{dn}{dt}$$

$$\boxed{\frac{1}{P}\frac{dP}{dt}} = \frac{1}{T}\frac{dT}{dt} + \frac{1}{n}\frac{dn}{dt}$$

$$\frac{dn}{dt} = \dot{n}_{vap}$$

$$\frac{dn}{dt} = \frac{n}{P}\frac{dP}{dt} - \frac{n}{T}\frac{dT}{dt}$$

Gas vaporizing from the liquid (H_2O)

$$PV = n^v RT$$

$$P_{sat} = e^{A - B/T}$$

$$\frac{1}{P}\frac{dP}{dt} = \frac{1}{T}\frac{dT}{dt} + \frac{1}{n_v}\frac{dn_v}{dt}$$

$$\frac{dP_{sat}}{dt} = + P_{sat}\frac{B}{T^2}\frac{dT}{dt}$$

$$\frac{dP}{dt} = \frac{P}{T}\frac{dT}{dt} + \frac{P}{n}\frac{dn^v}{dt}$$

$$\frac{P^{sat}}{T}\frac{dT}{dt} + \frac{P^{sat}}{n}\frac{dn}{dt} = + P_{sat}\frac{B}{T^2}\frac{dT}{dt}$$

$$\frac{P^{sat}}{n_{H_2O}}\frac{dn_{H_2O}}{dt} = \left[P_{sat}\frac{B}{T^2} - \frac{P_{sat}}{T}\right]\frac{dT}{dt}$$

5132 K = B

$$\frac{1}{n^v_{H_2O}}\frac{dn^v_{H_2O}}{dt} = \left[\frac{B}{T^2} - \frac{1}{T}\right]\frac{dT}{dt}$$

$$\frac{dn^v_{H_2O}}{dt} = n^v_{H_2O}\left[\frac{B}{T^2} - \frac{1}{T}\right]\frac{dT}{dt}$$

0.057 0.0032

Gas desorbing (CO_2) from the liquid

$$n_{des} = P\, m_L\, H$$

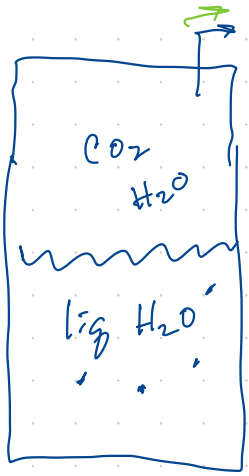
$$\frac{dn_{des}}{dt} = P m_L \frac{dH}{dt} + m_L H \frac{dP^{CO_2}}{dt}$$

$$\frac{dP^{CO_2}}{dt} = \frac{P}{n^v_{CO_2}} \frac{dn_{CO_2}}{dt} + \frac{P}{T} \frac{dT}{dt}$$

$$\frac{dn_{CO_2}}{dt} = P m_L \frac{dH}{dt} + m_L H \frac{P^{CO_2}}{n^v_{CO_2}} \cdot \frac{dn_{CO_2}}{dt} + \frac{m_L H P^{CO_2}}{T} \frac{dT}{dt}$$

$$P_{CO_2} = \frac{n^{CO_2} R T}{V_{gas}}$$

$$\frac{dn_{CO_2}}{dt}\left[1 - \frac{m_L H P}{n^v_{CO_2}}\right] = P^{CO_2} m_L \frac{dH}{dt} + \frac{m_L H P^{CO_2}}{T} \frac{dT}{dt}$$



How big should our relief valve be?

$\frac{dT}{dt}$ = Known

Q

fire heat flux

What is the flow rate of gas

$$\dot{n}_{out} = \frac{dn}{dt} = \frac{dn^{v}_{H_2O}}{dt} + \frac{dn^{v}_{CO_2}}{dt} =$$

???

$$n^{dis}_{CO_2} = m_L H P_{CO_2}$$

$$\dot{n}^{v}_{CO_2} = \frac{dn^{dis}_{CO_2}}{dt} = m_L H \frac{dP_{CO_2}}{dt} + m_L P_{CO_2} \frac{dH}{dt} + H P_{CO_2} \cancel{\frac{dm_L}{dt}}^{\,0}$$

$$\frac{dH}{dt} \propto \frac{dT}{dt} \leftarrow H = H_0 e^{-A\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

$$\frac{dH}{dt} = \frac{HA}{T^2}\frac{dT}{dt}$$

$\frac{dT}{dt}$

$$PV = nRT \qquad P^{CO_2} V_{gas} = n^{CO_2}_{gas} RT$$

$$\frac{1}{P^{CO_2}}\frac{dP^{CO_2}}{dt} = \frac{1}{n^{CO_2}_{g}}\frac{dn^{CO_2}}{dt} + \frac{1}{T}\frac{dT}{dt}$$

$$\frac{dP^{CO_2}}{dt} = \frac{P^{CO_2}}{n^{CO_2}_{g}}\frac{dn^{CO_2}}{dt} + \frac{P^{CO_2}}{T}\frac{dT}{dt}$$

$$P_{sat}^{H_2O} = e^{A - B/T}$$

August

$$\frac{dn_{H_2O}^{v}}{dt} = \dot{n}_{H_2O}^{v} = ?$$

$$\frac{dP_{sat}^{H_2O}}{dt} = e^{A - B/T} \cdot \frac{-1 \cdot -1}{T^2} \cdot B \cdot \frac{dT}{dt}$$

$$\frac{dP_{v}^{H_2O}}{dt} = e^{A - B/T} \cdot \frac{B}{T^2} \cdot \frac{dT}{dt} = P_{H_2O}^{sat} \cdot \frac{B}{T^2} \cdot \frac{dT}{dt}$$

$$\frac{1}{P_{v}^{H_2O}} \frac{dP_{v}^{H_2O}}{dt} = \frac{1}{n_{v}^{H_2O}} \frac{dn_{H_2O}^{vap}}{dt} + \frac{1}{T}\frac{dT}{dt}$$

$PV = nRT$ with constant V

$$\frac{P_{v}^{H_2O}}{n_{v}^{H_2O}} \frac{dn_{H_2O}^{vap}}{dt} + \frac{P_{v}^{H_2O}}{T}\frac{dT}{dt} = P_{H_2O}^{sat} \frac{B}{T^2}\frac{dT}{dt}$$

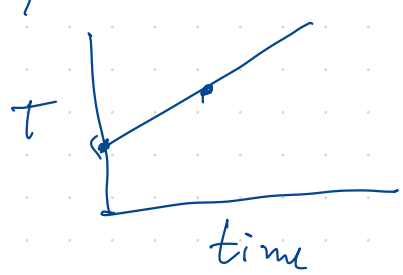
$\dot{n}_{H_2O}^{v}$

molar quantity of H_2O in gas

$$= \dot{n}_{H_2O}^{v} = \frac{\cancel{P_{H_2O}^{sat}}^{\,1}}{\cancel{P_{H_2O}^{v}}} \left[\frac{B}{T^2} - \frac{1}{T}\right] \frac{dT}{dt} \cdot n_{v}^{H_2O}$$

instantaneous change

flow into vapor from liquid water with increasing temp $\frac{dT}{dt}$



$$A = Be^{-C}$$

Take the derivative of A with respect to time. Assume all variable are functions of time.

$$\frac{dA}{dt} = B \cdot \frac{d}{dt}\left[e^{-C}\right] + e^{-C} \cdot \frac{d[B]}{dt}$$ chain rule

$$= Be^{-C} \cdot (-1) \cdot \frac{dC}{dt} + e^{-C} \frac{dB}{dt}$$